

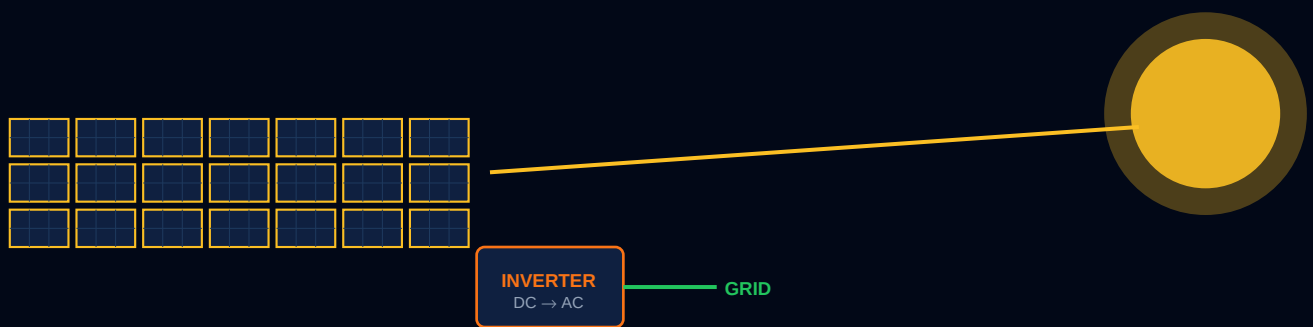


Fig. 1 — Grid-Tied Solar PV System Overview

SOLAR PV SYSTEMS

DESIGN GUIDE

Site Assessment
 Array Sizing
 Inverter Selection
 Battery Storage
 Grid Connection

Grid-Tied

System Types

Off-Grid

Battery Storage

Hybrid

Solutions

IEC/IEEE

Standards

Table of Contents

Chapter Overview

- Chapter 1 Solar PV Fundamentals — Physics, Technology & System Types
- Chapter 2 PV Module Technology & Specifications
- Chapter 3 Solar Resource Assessment & Site Analysis
- Chapter 4 Array Sizing & Yield Estimation
- Chapter 5 Grid-Tied System Design
- Chapter 6 String & Combiner Box Design
- Chapter 7 Battery Storage & Off-Grid Design
- Chapter 8 Inverter Selection & Configuration
- Chapter 9 Electrical Protection & Safety (IEC 60364-7-712)
- Chapter 10 Grid Connection & Utility Requirements
- Chapter 11 Monitoring, Commissioning & Performance
- Chapter 12 Economic Analysis & Financial Modelling

1

PV Fundamentals

Photovoltaic Physics, Technology & System Architectures

The Photovoltaic Effect

The photovoltaic effect is the generation of voltage and electric current in a material upon exposure to light. In a silicon solar cell, photons excite electrons across the p-n junction bandgap (1.12 eV for silicon), creating electron-hole pairs that are separated by the built-in electric field, producing a DC current.

- Bandgap Energy: Silicon absorbs photons with energy > 1.12 eV (wavelengths < 1100 nm)
- Photon Energy: $E = hf = hc/\lambda$ ($h = 6.626 \times 10^{-34}$ J·s, c = speed of light)
- Quantum Efficiency: % of incident photons that generate electron-hole pairs
- Fill Factor: $FF = P_{max} / (V_{oc} \times I_{sc})$ — quality indicator, typical 0.7–0.85
- Module Efficiency: Ratio of electrical output to incident solar irradiance $\times$ area

Solar Cell Technologies

Technology	Efficiency (%)	Temperature Coeff	Cost (\$/W)	Application
Monocrystalline Si (PERC)	20–24	-0.35%/°C	0.25–0.40	Residential, commercial
Polycrystalline Si	16–19	-0.40%/°C	0.20–0.35	Utility scale
HJT (Heterojunction)	22–26	-0.25%/°C	0.35–0.55	Premium residential
TOPCon (Tunnel Oxide)	22–25	-0.30%/°C	0.30–0.50	Commercial, utility
CdTe (Thin Film)	18–22	-0.25%/°C	0.15–0.25	Utility scale
Bifacial	22–27 (+10–20% rear)	—	+5–10% premium	Ground-mount, rooftop

2

Module Technology

Specifications, Ratings & Performance Modelling

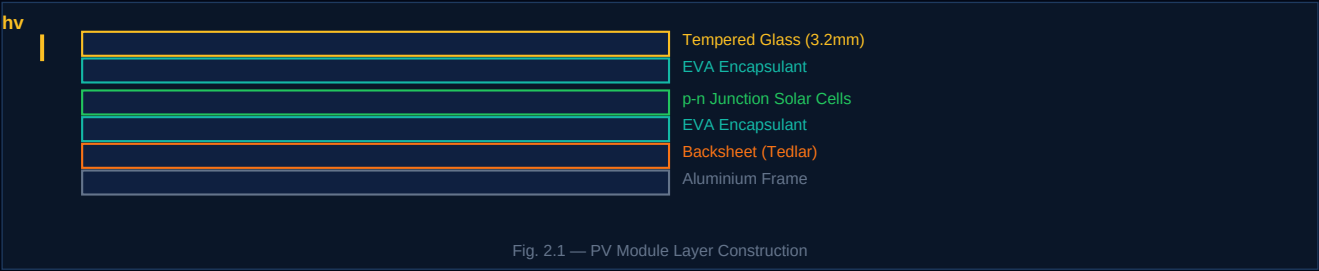
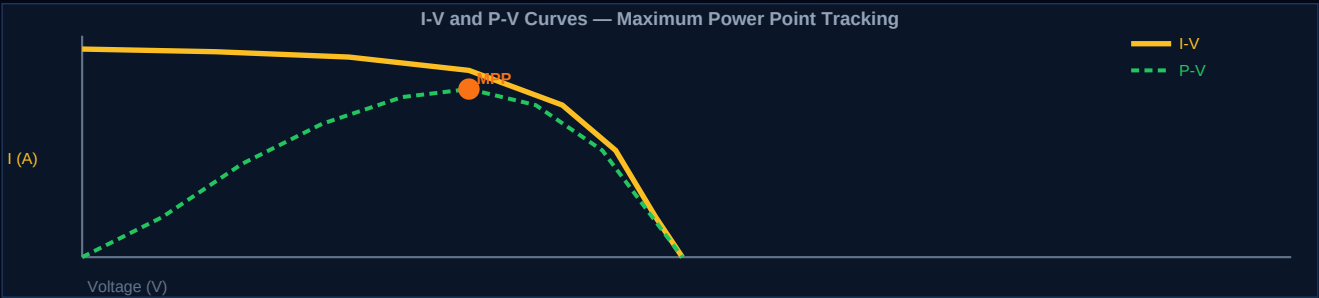


Fig. 2.1 — PV Module Layer Construction

Standard Test Conditions (STC)

STC: Irradiance = 1000 W/m² | Cell Temp = 25°C | AM 1.5 spectrum
 NOCT (Nominal Operating Cell Temp): 800 W/m², 20°C ambient, 1 m/s wind
 PTC (PV USA Test Conditions): 1000 W/m², 20°C ambient, 10 m/s wind — closer to real-world
 Temperature Correction: $P_{actual} = P_{STC} \times [1 + (T_c - 25) \times P_{max_coeff}]$



Key Module Parameters

- P_{max} (Wp): Peak power at STC — primary sizing parameter
- V_{oc} : Open-circuit voltage — determines max string voltage for inverter input
- I_{sc} : Short-circuit current — used for wire and fuse sizing
- V_{mp} / I_{mp} : Voltage/Current at maximum power point
- NOCT: Determines operating temperature for thermal correction
- Bifaciality Factor: Rear-side efficiency as % of front-side (typically 65–85%)

3

Solar Resource

Site Assessment, Irradiance Data & Shading Analysis

Solar Irradiance Components

- GHI (Global Horizontal Irradiance): Total irradiance on horizontal surface
- DNI (Direct Normal Irradiance): Beam radiation on surface perpendicular to sun
- DHI (Diffuse Horizontal Irradiance): Scattered radiation from sky (cloud/atmosphere)
- POA (Plane of Array): Irradiance on tilted module surface — what modules actually receive
- Peak Sun Hours (PSH): Hours/day of 1000 W/m² equivalent — same as kWh/m²/day

Irradiance Data Sources

Database	Source	Data Period	Resolution	Best Use
PVGIS	European Commission / JRC	1994–2022	Hourly/Monthly	European & global projects
NASA SSE / POWER	NASA Langley	1981–present	Monthly/Daily	Global, remote sites
SolarAnywhere	Clean Power Research	1998–present	30-min	North America
SolarGIS	GeoModel Solar	1994–present	15-min	High accuracy, global
Meteonorm	Meteotest	1991–2010	Hourly TMY	Simulation tools

Optimal Tilt & Orientation

- Rule of Thumb — Fixed Tilt: Tilt = Latitude (within 10-15° for most sites)
- South-facing (NH) or North-facing (SH) maximises annual yield
- East-West bifacial: +10-15% land use efficiency for utility sites
- Shading Loss Model: Use Solargis, PVSyst ShadeTool, or helioscope for accurate analysis
- Near Shading: Critical in first 1-2 hours after sunrise and before sunset

- Shade Analysis Tools: PVGIS (free), PVSyst, HelioScope, Aurora Solar
- Horizon Shading: Mountains, buildings — capture with Solmetric SunEye or drone survey
- Albedo: Ground reflectance affects bifacial yield — snow (0.8), concrete (0.2), grass (0.15)

4

Array Sizing

Energy Yield, System Losses & Performance Ratio

Array Sizing Methodology

- STEP 1: Determine annual energy target (kWh/yr) from load analysis
- STEP 2: Get annual PSH for site (kWh/m²/day) from solar database
- STEP 3: Array Size (kWp) = Annual Target / (365 × PSH × PR)
- PR (Performance Ratio): Typically 0.75–0.85 for well-designed system
- STEP 4: Number of Modules = Array Size (Wp) / Module P_max (Wp)

System Losses — PR Breakdown

Loss Source	Typical (%)	Best Practice (%)	Notes
Irradiance (spectral+angle)	2–4	2	Optical losses
Temperature	3–8	3	Depends on climate + module coeff
Shading	0–10	<2	Minimize with layout
Soiling & Dust	1–5	1–2	Clean regularly
Wiring & Transformer	1–3	1.5	Copper sizing
Inverter Efficiency	2–5	2	MPPT + conversion losses
Mismatch	1–3	1	Sort modules, use optimisers
Availability	0–3	1	Downtime for maintenance

Energy Yield Calculation Example

- Site: Addis Ababa, Ethiopia | PSH: 5.5 kWh/m²/day | PR: 0.80
- Annual Target: 50,000 kWh/yr (commercial building)
- Array Size = 50,000 / (365 × 5.5 × 0.80) = 31.2 kWp
- Modules @ 400 Wp each: 31,200 / 400 = 78 modules
- Roof Area Required: 78 × 2.0 m² = 156 m² (standard 400W module)
- Ground Coverage Ratio: 0.40–0.45 for row spacing with no shading at winter solstice

5

Grid-Tied Design

String Configuration, Inverter Matching & Grid Integration

String Sizing Fundamentals

- String Voltage: $V_{\text{string}} = N_{\text{modules}} \times V_{\text{mp}}$ (at operating temp)
- Max String Voltage: $N_{\text{max}} = V_{\text{inverter_max}} / V_{\text{oc_corrected}}$
- V_{oc} Temperature Correction: $V_{\text{oc_cold}} = V_{\text{oc_STC}} \times [1 + \beta \times (T_{\text{min}} - 25)]$
- Min String Voltage: Must keep array above inverter MPPT min voltage in heat
- $V_{\text{mp_hot}} = V_{\text{mp_STC}} \times [1 + \beta \times (T_{\text{max_cell}} - 25)]$ where β = temp coeff
- IEC 60364-7-712: Design for 40-year minimum design life — use voltage safety margin

STRING SIZING EXAMPLE:

Module: $V_{\text{oc}}=45.2\text{V}$, $V_{\text{mp}}=37.8\text{V}$, $\beta=-0.28\%/^{\circ}\text{C}$

Inverter MPPT: 200-800V range, max 1000V

$T_{\text{min}}=-5^{\circ}\text{C}$: $V_{\text{oc_cold}} = 45.2 \times [1 + (-0.0028) \times (-5 - 25)] = 45.2 \times 1.084 = 49.0\text{V}$

Max modules = $\text{floor}(1000 / 49.0) = 20$ modules per string

Min modules: $V_{\text{mp_hot}}$ at $70^{\circ}\text{C} = 37.8 \times [1 + (-0.0028) \times 45] = 37.8 \times 0.874 = 33.0\text{V}$

Min string = $\text{ceil}(200 / 33.0) = 7$ modules → typical string: 14-18 modules

Inverter Sizing — DC:AC Ratio

- DC:AC Ratio: Array kWp / Inverter kVA — typically 1.1–1.3
- Oversizing (clipping): Array output clipped when > inverter AC rating — intentional for economics
- Energy Clipping Loss: Calculate in PVsyst — accept <3% clipping for economics
- String Inverters: 1 MPPT per string group; best for uniform shading-free rooftops
- Central Inverters: >100kW; lower cost/W; requires combiner boxes; utility-scale
- Microinverters: 1 per module; shade tolerance; expensive; residential
- DC Optimisers: Module-level MPPT; shade tolerance; compatible with string inverters

6

String & Protection

Combiner Boxes, Cable Sizing & DC Protection

DC Cable Sizing (IEC 60364-7-712)

Cable Segment	Min Rating (A)	Temperature Rating	Typical Size
Module to string	$1.25 \times I_{sc}$	90°C (solar cable)	4–6 mm²
String to combiner	$1.25 \times I_{sc}$	90°C (solar cable)	4–10 mm²
Combiner to inverter	$1.25 \times \Sigma I_{sc}$	70°C (DC main)	10–95 mm²
Inverter AC output	Per inverter rating	70°C (AC main)	Per rating
Earth bonding	Min 6 mm² Cu	—	6–16 mm²

String Combiner Box Components

- String Fuses: $1.5 \times I_{sc}$ per string; IEC 60269-6 rated; positive AND negative conductor
- String DC Disconnect: Lockable isolator per IEC 60947-3; 1200V rated for 1000V systems
- Surge Protection (SPD): Type 2 at combiner box + inverter input; Type 1 if exposed site
- Monitoring CTs: Per-string current monitoring for fault detection
- Enclosure: IP65 minimum; anti-condensation heater for cold climates

VOLTAGE DROP LIMITS:

DC strings to inverter: Max 1.5% voltage drop (IEC 60364-7-712)

AC output to grid connection: Max 3% (local utility may require less)

Formula: $Vd\% = (2 \times L \times I \times \rho) / (A \times V_{nominal}) \times 100$

ρ (copper) = 0.0175 Ω·mm²/m at 20°C; derate for temperature

7

Battery Storage

Off-Grid Design, Battery Types & Charge Controllers

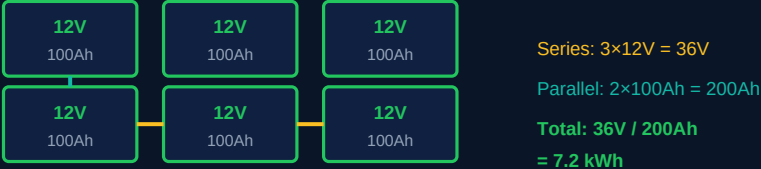


Fig. 7.1 — Battery Bank Series/Parallel Configuration

Battery Bank Series/Parallel Configuration

Battery Technologies for Solar Storage

Technology	Cycle Life	DoD (%)	Energy Density	Round-Trip Eff.	Cost (\$/kWh)
Flooded Lead-Acid	500–1500	50%	25–35 Wh/kg	70–80%	100–150
AGM/VRLA	500–1200	60%	30–45 Wh/kg	80–85%	150–250
LiFePO4 (LFP)	3000–6000	80–90%	90–160 Wh/kg	92–97%	200–350
NMC Li-ion	1000–3000	80%	150–220 Wh/kg	93–97%	150–300
Flow (Vanadium)	10,000+	100%	15–25 Wh/kg	65–75%	400–800

Battery Bank Sizing for Off-Grid Systems

- STEP 1: Daily Load (Wh/day) = Sum of [appliance W × hours/day]
- STEP 2: Autonomy Days (AD): Typically 1-3 days (no sun days)
- STEP 3: Required Capacity = Daily Load × AD / DoD
- STEP 4: Ah at system voltage = Required Wh / System Voltage (24V or 48V)
- STEP 5: Parallel strings = Required Ah / Battery Ah
- Example: 10kWh/day × 2 days / 0.8 DoD = 25 kWh → 25000Wh / 48V = 520 Ah

Charge Controller Selection

- PWM (Pulse Width Modulation): Simple, cheaper; array voltage must match battery voltage
- MPPT Charge Controller: Tracks maximum power point; 15–30% more efficient than PWM
- MPPT Input Range: Must accept string V_{oc} under cold conditions
- Sizing: Controller A rating ≥ 1.25 × I_{sc} of connected PV strings
- Temperature Compensation: Adjust charge voltage for battery temperature (typically -3mV/°C/cell)

8

Inverter Selection

Grid-Tied & Off-Grid Inverter Configuration

Grid-Tied Inverter Requirements

Requirement	Standard / Specification	Typical Range
Grid Voltage	IEC 62116 / EN 50549	230V / 400V (3-phase)
Frequency Tolerance	±1–2 Hz of nominal	49–51 Hz (50Hz grid)
THD (output)	<5% total harmonic distortion	< 3% premium inverters
Power Factor	Per utility requirements	0.9 lead – 0.9 lag
Anti-islanding	IEEE 1547 / IEC 62116	Certified required
Efficiency (Euro)	Weighted average	96–98.6% top inverters
DC Input Voltage	Model-specific range MPPT	200–1000 V typical

Inverter Types Comparison

- String Inverters (1–60 kW): Most common; simple; shade sensitivity; 1 per string group
- Central Inverters (>100 kW): Utility-scale; low cost/W; require combiner boxes
- Hybrid Inverters: Grid-tied + battery management in one unit; growing segment
- Off-Grid Inverter/Charger: Battery-based; creates its own grid; Victron, SMA Sunny Island
- Microinverters: Per-module AC; no DC arc hazard; premium cost; easy expansion

HYBRID INVERTER OPERATING MODES:

1. Grid + PV → Load (grid is priority backup)
2. PV → Load + Battery Charge (self-consumption priority)
3. Battery → Load (grid outage / peak-shaving)
4. Grid → Battery (time-of-use charging at low tariff)
5. Export to Grid (excess after load + battery full)

9

Protection & Safety

IEC 60364-7-712, Arc Fault & Earthing Design

DC Arc Fault Protection

- DC arc faults: Major fire cause in PV systems — series or parallel faults in DC wiring
- AFCI (Arc Fault Circuit Interrupter): Required by NEC 690.11 (US); growing adoption globally
- SMA / Fronius / Huawei: Most modern inverters include integrated arc detection
- Series arc: Broken conductor, corroded connector — detected by current signature
- Parallel arc: Insulation fault — detected by ground fault monitoring

Ground Fault & Insulation Monitoring

- Insulation Resistance: Measure at commissioning with 1000V megger; min 1 M Ω per string
- Ground Fault Detection: Inverter monitors for DC leakage to earth; trip on fault
- IT System (Floating): Both conductors ungrounded — IEC 60364-7-712 default; requires ISFL
- Earthing Conductors: Frame bonding min 4 mm² Cu; connect all frames to earth bus
- Lightning Protection: SPD at array (Type 1+2) and inverter (Type 2); earthing per IEC 62305

COMMISSIONING SAFETY CHECKS:

1. Verify polarity of each string before connecting to inverter
2. Insulation resistance test (all strings to earth) > 1 M Ω
3. V_{oc} measurement per string — verify matches design ($\pm 5\%$)
4. I_{sc} measurement with clamp meter — verify per string
5. Anti-islanding test: Disconnect grid, verify inverter shuts down in <5s

10

Grid Connection

Utility Requirements, Power Quality & Metering

Grid Connection Process

- Application: Submit technical specs (single-line diagram, equipment datasheets) to utility
- G99 / G98 (UK) / IEEE 1547 (US) / EN 50549 (EU): Local grid code compliance required
- Protection Relay: ROCOF (df/dt), voltage, frequency relays — type-tested per grid code
- Metering: Bidirectional smart meter for import/export measurement
- CT Sizing: Revenue-grade accuracy class 0.5 for grid connection point
- Export Limitation: Some utilities cap export (e.g., 50% of contracted capacity)

Power Quality Requirements

Parameter	Limit	Standard
THD (current)	< 5% at rated power	IEEE 1547 / EN 61000-3-2
Voltage imbalance	< 2% at PCC	EN 50160
Flicker (Pst)	< 0.35 / 0.25 (Plt)	EN 61000-3-3
DC injection	< 0.5% rated AC current	IEEE 1547
Power factor	> 0.95 lag–lead at rated power	Local grid code
Inrush current	Soft-start ramp rate limit	Utility specific

NET METERING & FEED-IN TARIFF:

Net Metering: Exported units offset imported units (1:1 credit)

FIT (Feed-in Tariff): Fixed price paid per kWh exported (utility-set rate)

Self-Consumption: Maximise by matching PV peak to load peak — add battery + DCS

Power Purchase Agreement (PPA): Developer funds system; building owner buys power at discount

11

Monitoring & Commissioning

Performance Analysis, Fault Detection & O&M

Monitoring System Architecture

- String-Level Monitoring: Current per string (combiner box CTs) — detects shading, faults
- Module-Level Monitoring: Microinverter or optimiser data — full visibility, highest cost
- Weather Station: Irradiance (pyranometer), temperature, wind — required for PR calculation
- Communication: Inverter RS485/Modbus → datalogger → cloud platform
- Alerts: Set alarms for string underperformance (>10% deviation from expected)

Key Performance Indicators (KPIs)

KPI	Formula	Good Value	Poor Value
Performance Ratio (PR)	$E_{\text{actual}} / (G_{\text{POA}} \times P_{\text{STC}})$	> 80%	< 70%
Capacity Factor	$E_{\text{annual}} / (P_{\text{STC}} \times 8760\text{h})$	15–25% (utility)	< 12%
Specific Yield	$E_{\text{annual}} \text{ (kWh)} / P_{\text{STC}} \text{ (kWp)}$	900–1800 kWh/kWp	< 700
System Availability	Operating hours / Total hours	> 99%	< 95%
Energy Clipping	Clipped kWh / Yield kWh	< 3%	> 5%

- FAULT DETECTION PRIORITY:
1. Zero output string: Check string fuse, connector, module damage
 2. Low output (30–70%): Partial shading, soiling, module degradation
 3. High temperature shutdown: Check ventilation, derating setting
 4. Grid fault: Check AC voltage, frequency, anti-islanding relay
 5. Insulation fault: Megger test all strings to earth; check connectors

12 Financial Analysis

LCOE, Payback, IRR & Project Financing

Key Financial Metrics

- **LCOE (Levelised Cost of Energy):** Total lifecycle cost / Total lifetime kWh produced
- **Simple Payback:** System cost / Annual savings (years) — 4–8 years typical
- **NPV (Net Present Value):** PV of future cash flows minus initial cost
- **IRR (Internal Rate of Return):** Discount rate that makes NPV = 0; compare to WACC
- **Degradation Rate:** Module output degrades ~0.5–0.7%/year (linear); account over 25 years

LCOE Calculation Framework

$$\text{LCOE (\$/kWh)} = (\text{CAPEX} + \text{PV of O\&M} + \text{PV of replacement}) / \text{PV of lifetime yield}$$

Global Benchmarks (2025):

- Utility-scale ground mount: \$0.020–0.045 / kWh
- Commercial rooftop C&I: \$0.040–0.080 / kWh
- Residential rooftop: \$0.080–0.150 / kWh
- Off-grid with battery: \$0.150–0.300 / kWh

Financing Options

Finance Model	Description	Who Bears Risk	Typical IRR
Cash Purchase	Owner pays upfront	Owner	8–15%
Bank Loan (secured)	Debt financing; own asset	Owner	10–20% equity
Operating Lease	Pay monthly; off-balance sheet	Developer	—
PPA	Buy power, not system	Developer (owns)	8–12%
Green Bond	Corporate bond for RE projects	Issuer	Bond rate + margin

- **O&M; Cost:** Budget \$10–15/kWp/year (cleaning, monitoring, insurance, inverter maintenance)
- **Insurance:** Property + liability; specify in CAPEX budget (typically 0.25–0.5% of system cost/yr)
- **Inverter Replacement:** Budget for 1× inverter replacement at year 12–15 (string inverters)